Generalized Proof of Theorem 3.3

Event-Based Ordering Induces Local Information
under General Compliance (Definition 3.1)

Rebuttal Supplement — Corrected Version

1 Setup

We restate the relevant definitions from the paper for self-containedness.

Definition 1 (Alphabet and Key Subset)**.** *Let $\mathcal{A}$ be a finite alphabet with $|\mathcal{A}| = \ell \geq 2$. Let $S \subseteq \mathcal{A}$ with $|S| = m \geq 2$, and let $\kappa : S \to \{1, \dots, m\}$ be a bijection defining a total order on S. Let $X = (X_1, \dots, X_n)$ with $X_t \overset{i.i.d.}{\sim} \mathrm{Uniform}(\mathcal{A})$.*

Definition 2 (General Compliance — Definition 3.1 of the paper)**.** *For lag parameter $\lambda \geq 1$, the sequence X is* compliant *if there exists a chain of positions $(t_1, \dots, t_k)$ with $k \geq 2$ such that:*

(i) $t_{j+1} - t_j = \lambda$ *for all* j,

(ii) $X_{t_j} \in S$ *for all* j, *and*

(iii) $\kappa(X_{t_1}) \leq \cdots \leq \kappa(X_{t_k})$.

Define the compliance indicator $Y^* = \mathbf{1}\{X \text{ is compliant}\}$ and the compliance probability $\rho = P(Y^* = 1)$.

Remark 3. *The proof in Appendix C.1 of our submission establishes Theorem 3.3 for the* narrow *(naive) case where $S = \mathcal{A}$, $\lambda = 1$, and κ is the natural order, using a stars-and-bars counting argument. As the reviewer correctly noted, the general case (arbitrary S, κ, λ) requires a separate argument. We provide this below.*

2 Generalized Theorem and Proof

Theorem 4 (Event-Based Ordering Induces Local Information — General Case)**.** *Under Definition 2 with $|\mathcal{A}| = \ell \geq 2$, $|S| = m \geq 2$, any bijection κ, any $\lambda \geq 1$, $n \geq \lambda + 1$, and compliance probability $\rho \in (0, 1)$, we have*

$$I(Y^*; X_1) > 0.$$

Remark 5 (Non-degeneracy condition $\rho \in (0,1)$)**.** *The condition $\rho \in (0,1)$ excludes degenerate cases where the classification task is trivial (Y^* is constant).*

When $\rho = 0$: this occurs only when $n < \lambda + 1$ (no chain of length ≥ 2 can exist), which is excluded by assumption.

When $\rho = 1$: this can occur for $S = \mathcal{A}$ when the sequence is long relative to both λ and ℓ. Specifically, a sequence is non-compliant only if, within every *residue class modulo λ, consecutive values at distance λ are strictly κ-decreasing. The longest residue class has $\lceil n/\lambda \rceil$ elements, so* *strictly decreasing sequences exist only if $\lceil n/\lambda \rceil \le \ell$. When $\lceil n/\lambda \rceil > \ell$ and $S = \mathcal{A}$, we get $\rho = 1$ (every sequence is compliant and there is nothing to classify). This case is of no practical interest.*

When $S \subsetneq \mathcal{A}$ and $n \ge \lambda + 1$: we always have $\rho \in (0, 1)$. Indeed, $\rho > 0$ because we can construct a compliant sequence (place two key letters in non-decreasing κ-order at distance λ), and $\rho < 1$ because the all-non-key sequence ($X_t \notin S$ for all t, which has positive probability since $|\mathcal{A} \setminus S| \ge 1$) is non-compliant.

In the paper's experimental settings ($S \subsetneq \mathcal{A}$ for the tricky and parity tasks; $S = \mathcal{A}$ with $n = 20$, $\ell = 26$, $\lambda = 1$ for the naive task giving $\lceil 20/1 \rceil = 20 \le 26$), the non-degeneracy condition is always satisfied.

Proof of Theorem 4. It suffices to show that $P(Y^* = 1 \mid X_1 = a)$ depends on the choice of symbol a, since this implies $P(Y^* = 1, X_1 = a) \ne P(Y^* = 1)\, P(X_1 = a)$ for some a, and hence $I(Y^*; X_1) > 0$.

We treat two cases.

Case 1: $S \subsetneq \mathcal{A}$ (there exist non-key symbols).

Let $s_{\min} \in S$ satisfy $\kappa(s_{\min}) = 1$ (minimum rank), and let $a \in \mathcal{A} \setminus S$ be any non-key symbol. We show $P(Y^* = 1 \mid X_1 = s_{\min}) > P(Y^* = 1 \mid X_1 = a)$ via a coupling argument.

Coupling construction. Consider two sequences $\mathbf{X}$ and $\mathbf{X}'$ that are identical at all positions $t \ge 2$, but differ at position 1:

$$X_1 = s_{\min} \in S, \qquad X'_1 = a \notin S, \qquad X_t = X'_t \text{ for all } t \ge 2.$$

Since entries are i.i.d. Uniform($\mathcal{A}$), this coupling is valid: both $\mathbf{X}$ (conditioned on $X_1 = s_{\min}$) and $\mathbf{X}'$ (conditioned on $X'_1 = a$) have the correct conditional distributions over positions ≥ 2.

Step 1: Compliance of $\mathbf{X}'$ implies compliance of $\mathbf{X}$. Suppose $\mathbf{X}'$ is compliant via chain $(t_1, \ldots, t_k)$. Compliance requires $X'_{t_j} \in S$ for every j. Since $X'_1 = a \notin S$, no chain element can be at position 1, so $t_j \ge 2$ for all j. At these positions $X_{t_j} = X'_{t_j}$, so the same chain makes $\mathbf{X}$ compliant:

$$\{\mathbf{X}' \text{ is compliant}\} \;\subseteq\; \{\mathbf{X} \text{ is compliant}\}.$$

Step 2: The inclusion is strict with positive probability. Since $n \ge \lambda + 1$, position $1 + \lambda$ exists. Choose the following realization of $(X_2, \ldots, X_n)$:

- $X_{1+\lambda} = s$ for some $s \in S$ (so $\kappa(s) \ge 1$),
- $X_t \notin S$ for all $t \ge 2$ with $t \ne 1 + \lambda$.

This has positive probability since $|\mathcal{A} \setminus S| \ge 1$. Under this realization:

- $\mathbf{X}$ is compliant: the chain $(1,\ 1{+}\lambda)$ satisfies $X_1 = s_{\min} \in S$, $X_{1+\lambda} = s \in S$, and $\kappa(s_{\min}) = 1 \le \kappa(s)$.
- $\mathbf{X}'$ is *not* compliant: position 1 cannot participate in any chain ($X'_1 = a \notin S$), and among positions ≥ 2, the unique key symbol is s at position $1{+}\lambda$, which alone cannot form a chain of length ≥ 2.

Therefore:

$$P(Y^* = 1 \mid X_1 = s_{\min}) \;>\; P(Y^* = 1 \mid X_1 = a).$$

Case 2: $S = \mathcal{A}$ (all symbols are key), with $\rho < 1$.

This case admits a direct argument without coupling.

Let $s_{\min} \in S$ satisfy $\kappa(s_{\min}) = 1$. Since $n \ge \lambda + 1$, positions 1 and $1 + \lambda$ both exist. For *any* realization of $(X_2, \ldots, X_n)$, consider the chain $(1,\ 1{+}\lambda)$: we have $X_1 = s_{\min} \in S = \mathcal{A}$ and $X_{1+\lambda} \in \mathcal{A} = S$, with

$$\kappa(X_1) = \kappa(s_{\min}) = 1 \;\le\; \kappa(X_{1+\lambda}).$$

The last inequality holds for every possible value of $X_{1+\lambda}$, since 1 is the minimum rank. Therefore the chain $(1, 1+\lambda)$ is *always* compliant, regardless of the rest of the sequence:

$$P(Y^* = 1 \mid X_1 = s_{\min}) \;=\; 1.$$

Since $\rho = P(Y^* = 1) < 1$ by assumption, we have $P(Y^* = 0) > 0$. Because $P(Y^* = 0 \mid X_1 = s_{\min}) = 0$, the total probability formula

$$P(Y^* = 0) \;=\; \sum_{a \in \mathcal{A}} P(Y^* = 0 \mid X_1 = a)\, P(X_1 = a) \;=\; \frac{1}{\ell} \sum_{a \in \mathcal{A}} P(Y^* = 0 \mid X_1 = a)$$

implies that there exists at least one symbol $a^* \in \mathcal{A}$ with $P(Y^* = 0 \mid X_1 = a^*) > 0$, i.e., $P(Y^* = 1 \mid X_1 = a^*) < 1$. Therefore:

$$P(Y^* = 1 \mid X_1 = s_{\min}) = 1 \;>\; P(Y^* = 1 \mid X_1 = a^*).$$

Conclusion. In both cases, $P(Y^* = 1 \mid X_1)$ is non-constant, and therefore $I(Y^*; X_1) > 0$. $\square$

3 Remarks

Remark 6 (Relation to the Narrow Case)**.** *The stars-and-bars proof in Appendix C.1 provides an* exact computation *of $P(X_1 = a_j \mid Y^* = 1)$ for the narrow setting ($S = \mathcal{A}$, $\lambda = 1$), yielding a closed-form expression for the conditional distribution and hence the mutual information. The proof above is less quantitative but applies to the full generality of Definition 3.1. The two are complementary: the narrow case gives precise magnitudes; the general case establishes that local information leakage is* unavoidable *for any non-trivial ordering-based compliance task.*

Remark 7 (Case 2 admits a stronger characterization)**.** *In Case 2 ($S = \mathcal{A}$), one can further show that $P(Y^* = 1 \mid X_1 = a)$ is non-increasing in $\kappa(a)$. Intuitively, having a lower-ranked symbol at position 1 makes it easier to start compliant chains (the non-decreasing condition $\kappa(X_1) \le \kappa(X_{1+\lambda})$ becomes less restrictive). This can be formalized via the same coupling used in Case 1: for $\kappa(a) < \kappa(a')$, couple $\mathbf{X}$ ($X_1 = a$) with $\mathbf{X}'$ ($X_1 = a'$) at all positions ≥ 2; any compliant chain starting at position 1 in $\mathbf{X}'$ requires $\kappa(X_{t_2}) \ge \kappa(a') > \kappa(a)$, and replacing a' with a only relaxes this constraint. Chains not involving position 1 are identical. Hence $P(Y^* = 1 \mid X_1 = a) \ge P(Y^* = 1 \mid X_1 = a')$.*

4 Condensed Proof for the Paper

For space-efficient inclusion in a revised manuscript, we provide the following condensed version.

Theorem 8 (Event-Based Ordering Induces Local Information — General Case)**.** *Under Definition 3.1 with $|S| = m \ge 2$, any κ, any $\lambda \ge 1$, $n \ge \lambda + 1$, and $\rho \in (0, 1)$: $I(Y^*; X_1) > 0$.*

Proof. Let $s_{\min} \in S$ with $\kappa(s_{\min}) = 1$.

Case $S \subsetneq \mathcal{A}$. Fix $a \in \mathcal{A} \setminus S$ and couple sequences $\mathbf{X}$ and $\mathbf{X}'$, identical at positions $2, \ldots, n$, with $X_1 = s_{\min}$ and $X_1' = a$. Since $a \notin S$, every compliant chain in $\mathbf{X}'$ avoids position 1 and remains valid in $\mathbf{X}$, so $\{\mathbf{X}' \text{ compl.}\} \subseteq \{\mathbf{X} \text{ compl.}\}$. Conversely, setting $X_{1+\lambda} \in S$ with all other positions ≥ 2 outside S (positive probability since $S \subsetneq \mathcal{A}$) makes $\mathbf{X}$ compliant via chain $(1, 1+\lambda)$ while $\mathbf{X}'$ is not. Hence $P(Y^* = 1 \mid X_1 = s_{\min}) > P(Y^* = 1 \mid X_1 = a)$.

Case $S = \mathcal{A}$, $\rho < 1$. Since $S = \mathcal{A}$, every symbol has a κ-rank. For any realization, $X_1 = s_{\min}$ and $X_{1+\lambda} \in S$ satisfy $\kappa(s_{\min}) = 1 \le \kappa(X_{1+\lambda})$, so chain $(1, 1+\lambda)$ is always compliant: $P(Y^* = 1 \mid X_1 = s_{\min}) = 1$. Since $\rho < 1$, the law of total probability implies there exists a^* with $P(Y^* = 1 \mid X_1 = a^*) < 1$.

In both cases $P(Y^* = 1 \mid X_1)$ is non-constant, so $I(Y^*; X_1) > 0$. $\square$